



School of Future Tech

Case Study Report

ON

**BioGazer – DNA Analysis and Mutation
Tracking System**

BY

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Problem Statement :-

Modern genetic research involves analyzing large DNA sequences and identifying mutations efficiently. Researchers require a platform that can visualize DNA sequences, search genetic information, manage patient records, and assess mutation risks through an interactive interface.

BioGazer is a React.js-based DNA Analysis and Visualization System that enables scientists to view, edit, validate, and analyze genetic sequences. The platform provides color-coded DNA visualization, mutation tracking, gene searching, risk analysis, patient record management, and customizable settings through a modern dashboard interface.

Tech Stack :-

Frontend Technologies

- React.js
- JavaScript
- HTML
- CSS

Libraries and Tools

- React Router DOM
- Axios
- Vite

React Concepts Used

- Functional Components
- useState
- useEffect
- useContext
- useMemo

- useCallback
- Props
- Conditional Rendering
- Lists and Keys
- Context API

Development Tools

- Visual Studio Code
- GitHub
- Vercel

Component Architecture :-

The project follows a component-based architecture to improve scalability, maintainability, and reusability.

Main Components

- **Navbar**
- **Sidebar**
- **Layout**
- **StatCard**
- **DNA Strand Viewer**
- **Patient Table**
- **Pagination**
- **Risk Badge**
- **Loader**
- **Empty State**

Pages

- **Home Dashboard**
- **DNA Viewer**
- **Patients**
- **Gene Search**
- **Risk Calculator**

- **Settings**

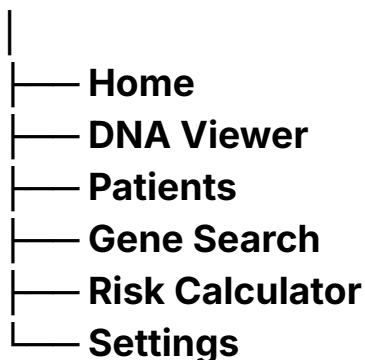
Context Management

AppContext is used to manage:

- **Theme Settings**
- **Notification Settings**
- **DNA Display Settings**
- **Global Application State**

Routing Structure

App Component



Features :-

1. Dashboard

Displays project statistics, DNA information, and quick navigation options.

2. Color-Coded DNA Viewer

Visualizes DNA sequences using different colors for A, T, G, and C nucleotides.

3. DNA Mutation Tracking

Identifies and highlights mutations in DNA sequences.

4. Gene Editing with Undo/Redo

Allows users to edit DNA sequences and restore previous changes.

5. DNA Sequence Organizer

Provides search, sorting, and filtering of DNA records.

6. Patient Records Management

Manages and displays 1200+ patient records efficiently.

7. Mutation Risk Calculator

Calculates and displays mutation risk levels based on mutation count.

8. Smart Gene Search

Performs real-time searching and filtering of gene information.

9. DNA Validation System

Validates DNA input and accepts only valid nucleotide characters.

10. Study Settings Hub

Allows users to customize themes, notifications, and application settings.

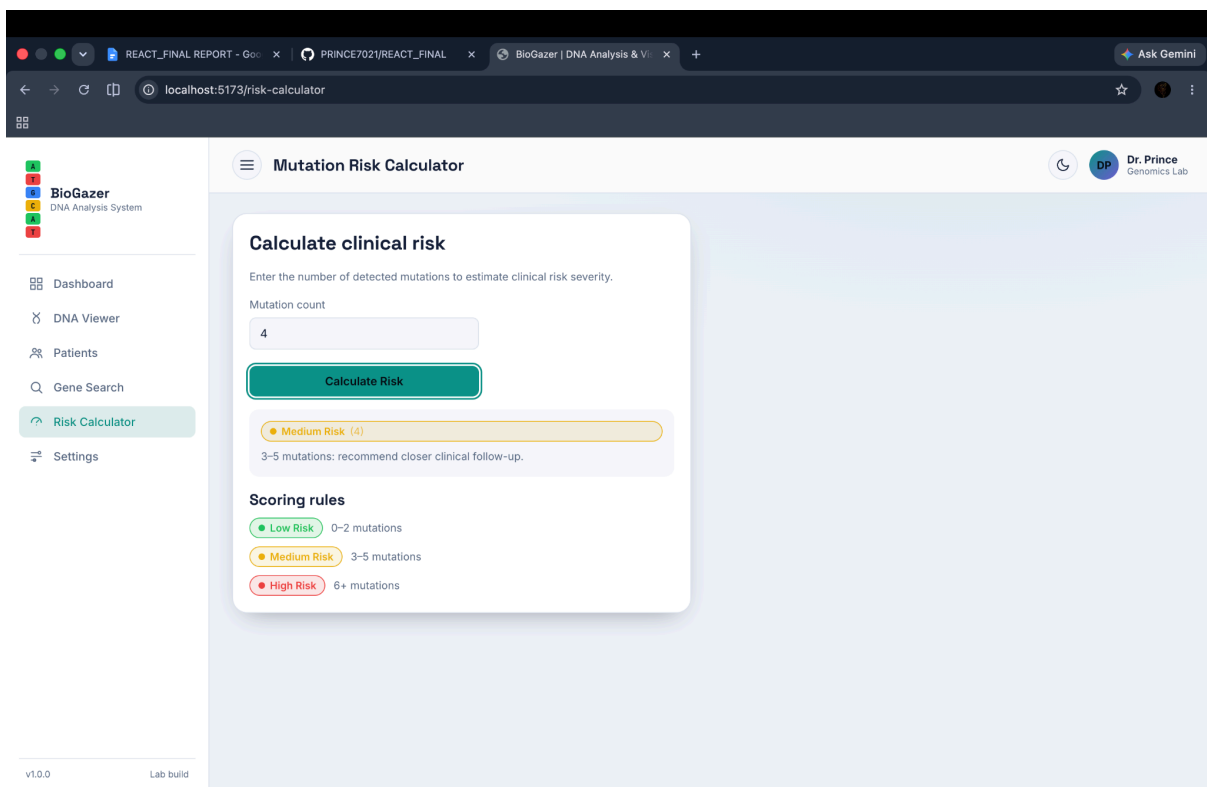
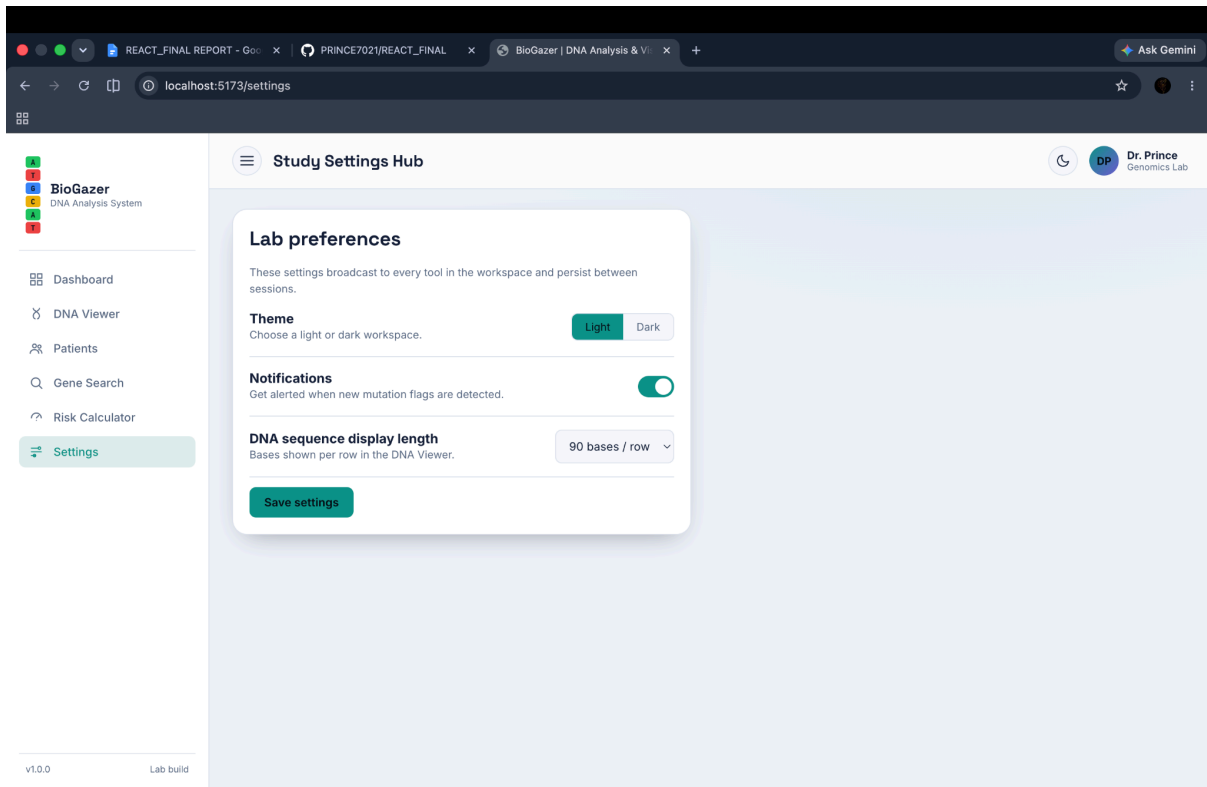
11. Performance Optimization

Uses pagination and React optimization techniques for smooth performance.

12. Responsive Design

Provides a user-friendly interface across desktop, tablet, and mobile devices.

Screenshots :-



BioGazer

DNA Analysis System

Dashboard

DNA Viewer

Patients

Gene Search

Risk Calculator

Settings

v1.0.0

Lab build

Smart Gene Search

Search the gene reference index10 matches

Try BRCA1, TP53, KRAS...

BRCA1

Breast Cancer gene 1

Tumor suppressor gene; pathogenic variants raise hereditary breast and ovarian cancer risk.

BRCA2

Breast Cancer gene 2

DNA repair gene; mutations are linked to hereditary breast, ovarian and pancreatic cancers.

TP53

Tumor Protein P53

The "guardian of the genome"; mutated in a majority of human cancers including Li-Fraumeni syndrome.

EGFR

Epidermal Growth Factor Receptor

Cell-surface receptor; activating mutations drive non-small-cell lung cancer growth.

KRAS

Kirsten Rat Sarcoma viral oncogene

PTEN

Phosphatase and Tensin

Gene Detail

BRCA1

Breast Cancer gene 1

CHROMOSOME17q21.31

SUMMARY

This gene encodes a 190 kD nuclear phosphoprotein that plays a role in maintaining genomic stability, and it also acts as a tumor suppressor. The BRCA1 gene contains 22 exons spanning about 110 kb of DNA. The encoded protein combines with other tumor suppressors, DNA damage sensors, and signal transducers to form a large multi-subunit protein complex known as the BRCA1-associated genome surveillance complex (BASC). This gene product associates with RNA polymerase II, and through the C-terminal domain, also interacts with histone deacetylase complexes. This protein thus plays a role in transcription, DNA repair of double-stranded breaks, and recombination. Mutations in this gene are responsible for approximately 40% of inherited breast cancers and more than 80% of inherited breast and ovarian cancers. Alternative splicing plays a role in modulating the subcellular localization and physiological function of this gene. Many alternatively spliced transcript variants, some of which are disease-associated mutations, have been described for this gene, but the full-length natures of only some of these variants has been described. A related pseudogene, which is also located on chromosome 17, has been identified. [provided by RefSeq, May 2020].

SOURCE

Live: MyGene.info

BioGazer

DNA Analysis System

Dashboard

DNA Viewer

Patients

Gene Search

Risk Calculator

Settings

v1.0.0

Lab build

Patient Records

Patient Registry1,200 of 1,200

Search by name, patient ID or DNA ID...

All risk levels

All genders

Sort: Patient ID

Patient ID	Name	Age	Gender	DNA ID	Mutations	Risk Level
PT-00001	Dev Verma	81	Other	SEQ-0022	2	Low Risk
PT-00002	Dev Walker	45	Female	SEQ-0030	10	High Risk
PT-00003	Ananya Brown	66	Male	SEQ-0019	0	Low Risk
PT-00004	Vivian Clark	3	Female	SEQ-0008	0	Low Risk
PT-00005	Emma Sharma	17	Other	SEQ-0033	5	Medium Risk
PT-00006	Riya Sharma	5	Male	SEQ-0010	7	High Risk
PT-00007	Ethan Iyer	71	Other	SEQ-0009	6	High Risk
PT-00008	Kabir Patel	63	Other	SEQ-0038	8	High Risk
PT-00009	Karan Verma	90	Female	SEQ-0001	1	Low Risk
PT-00010	Lucas Garcia	86	Female	SEQ-0038	8	High Risk
PT-00011	Maya Mehta	50	Other	SEQ-0040	1	Low Risk
PT-00012	Nora Verma	55	Male	SEQ-0013	7	High Risk

Conclusion :-

BioGazer successfully demonstrates the use of React JS in developing a modern scientific web application. The project provides an effective platform for DNA visualization, mutation tracking, gene searching, and risk analysis. Through reusable React components and a responsive user interface, the application improves the accessibility and management of genetic data while showcasing modern frontend development practices.

GitHub Repository Link :-

https://github.com/PRINCE7021/REACT_FINAL

Live Project Link :-